Capital Expansion Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Expansion Intensity (CEI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on CEI achieves an annualized gross (net) Sharpe ratio of 0.46 (0.37), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 30 (23) bps/month with a t-statistic of 4.04 (3.22), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets) is 24 bps/month with a t-statistic of 3.34.

1 Introduction

Production-based asset pricing theory posits that firm characteristics related to investment and production decisions contain information about expected returns through their connection to firms' marginal costs of investment and exposure to aggregate productivity shocks. While a substantial literature documents that various measures of corporate investment predict cross-sectional returns, the economic mechanisms linking specific forms of capital expansion to risk premia remain incompletely understood. We examine Capital Expansion Intensity (CEI), a comprehensive measure that captures firms' rate of change in productive capacity relative to their existing capital base, to understand how production dynamics shape the cross-section of expected returns.

In a production economy, firms' investment decisions reflect the first-order conditions that equate the marginal cost of investment to the present value of future marginal products of capital (Cochrane, 1991; Zhang, 2005). Capital Expansion Intensity directly measures the rate at which firms adjust their productive capacity, thereby proxying for both the marginal costs of investment and firms' exposure to productivity shocks. When firms undertake rapid capital expansion, they face increasing marginal adjustment costs that reduce the covariance between their cash flows and the stochastic discount factor, leading to lower required returns (Cooper and Ejarque, 2006; Liu et al., 2009).

The production-based framework predicts that firms with high CEI experience diminishing marginal products of capital due to decreasing returns to scale and convex adjustment costs. These frictions create a wedge between the marginal and average products of capital that varies systematically with the intensity of capital expansion (Bala and Nagel, 2020). Moreover, high CEI firms have already exercised valuable growth options, reducing their exposure to aggregate productivity risk and lowering their conditional risk premia (Carlson et al., 2004).

Cross-sectional variation in CEI also captures heterogeneity in firms' exposure to investment-specific technology shocks, a key source of aggregate risk in production economies (Papanikolaou, 2011; Kogan and Papanikolaou, 2014). Firms undertaking intensive capital expansion have typically adopted the latest vintage of capital goods, making them less sensitive to future improvements in investment technology. This reduced exposure to technology shocks translates into lower expected returns, as these firms provide less valuable hedging opportunities against aggregate productivity fluctuations.

Our empirical analysis reveals that Capital Expansion Intensity strongly predicts cross-sectional returns. A value-weighted long/short trading strategy based on CEI achieves an annualized gross Sharpe ratio of 0.46, with a monthly average excess return of 25 basis points (t-statistic = 3.57). The strategy's performance remains robust after controlling for established risk factors, generating a monthly alpha of 30 basis points (t-statistic = 4.04) relative to the Fama and French (2018) six-factor model.

The economic magnitude of the CEI effect is substantial and pervasive across different segments of the market. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the CEI strategy delivers an average return of 22 basis points per month (t-statistic = 2.45), with six-factor alphas ranging from 24 to 29 basis points. These results demonstrate that the predictive power of CEI is not confined to small, illiquid stocks where limits to arbitrage might prevent efficient pricing.

Accounting for transaction costs using the composite bid-ask spread measure of Chen and Velikov (2022), the CEI strategy maintains strong performance with a net Sharpe ratio of 0.37 and net six-factor alpha of 23 basis points per month (t-statistic = 3.22). The strategy's gross monthly alpha remains significant at 24 basis points (t-statistic = 3.34) even after controlling for the six Fama-French factors plus six closely related anomalies including growth in book equity, inventory growth, and changes in net operating assets. These results establish CEI as an economically important and distinct source of cross-sectional return variation.

This paper contributes to the production-based asset pricing literature by establishing Capital Expansion Intensity as a powerful predictor of cross-sectional returns that captures unique information about firms' exposure to productivity and investment risks. While Cooper et al. (2008) document that asset growth predicts returns and Lyandres et al. (2019) show that capital investment relates to expected returns, our CEI measure provides a more comprehensive characterization of firms' production dynamics by capturing the intensive margin of capital adjustment. Unlike the investment factor in Fama and French (2015) that focuses on changes in total assets, CEI specifically isolates variation in productive capacity expansion that theory links to marginal costs and conditional risk premia.

Our findings extend the investment-based anomaly literature documented in Titman et al. (2004) and Xing (2008) by demonstrating that CEI contains incremental information beyond established investment and growth measures. The CEI strategy's alpha of 24 basis points remains highly significant after controlling for closely related anomalies, indicating that it captures a distinct dimension of production-related risk. This result suggests that existing factor models incompletely capture the risk-return tradeoffs embedded in firms' capital expansion decisions, consistent with the theoretical predictions of Zhang (2005) regarding the importance of investment frictions in asset pricing.

The robust performance of CEI across size quintiles and after transaction costs provides new evidence supporting production-based theories of the cross-section. Our results demonstrate that production characteristics matter for asset prices not through behavioral biases or limits to arbitrage, but because they reflect fundamental differences in firms' risk exposures arising from optimal investment decisions under uncertainty. These findings reinforce the importance of incorporating production-side frictions and technology shocks into asset pricing models, as emphasized by Kogan and Papanikolaou (2013) and Garlappi and Song (2017), and suggest that Capital Expansion Intensity serves as a valuable empirical proxy for these theoretical constructs.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. The sample spans multiple decades, allowing us to examine the Capital Expansion Intensity signal across various market conditions and business cycles.Capital Expansion Intensity signal relies on two key accounting variables. Invested capital (ICAPT) represents the total capital invested in a company's operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This measure captures the cumulative funding deployed in the firm's productive assets and operations. Nonoperating income-other (NOPIO) comprises income and expenses from activities not directly related to the firm's core operations, including items such as restructuring charges, litigation settlements, asset write-downs, and other special items that fall outside regular business activities.construct the Capital Expansion Intensity signal by computing the year-over-year change in invested capital, scaled by lagged nonoperating income-other: (ICAPT(t) - ICAPT(t-1)) / NOPIO(t-1). This ratio captures the relationship between a firm's capital expansion rate and its nonoperating activities from the previous period. The economic intuition is that scaling capital growth by nonoperating income provides insight into how firms fund expansion through non-core activities and special items. We calculate the signal using fiscal year-end values and require non-missing values for both ICAPT and NOPIO. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative values of NOPIO in the denominator are excluded from the analysis to ensure meaningful ratio calculations.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CEI signal. Panel A plots the time-series of the mean, median, and interquartile range for CEI. On average, the cross-sectional mean (median) CEI is -13.21 (-3.43) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CEI data. The signal's interquartile range spans -41.48 to 30.08. Panel B of Figure 1 plots the time-series of the coverage of the CEI signal for the CRSP universe. On average, the CEI signal is available for 4.80% of CRSP names, which on average make up 6.77% of total market capitalization.

4 Does CEI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CEI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CEI portfolio and sells the low CEI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CEI strategy earns an average return of 0.25% per month with a t-statistic of 3.57. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.26% to 0.30% per month and have t-statistics exceeding 3.62 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.14, with a t-statistic of -3.98 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 441 stocks and an average market capitalization of at least $1,516 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 2.70. Out of the twenty-five alphas reported in

Panel A, the t-statistics for twenty-five exceed two, and for twenty-two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 14-21bps/month. The lowest return, (14 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.11. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CEI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the CEI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CEI, as well as average returns and alphas for long/short trading CEI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the CEI strategy achieves an average return of 22 bps/month with a t-statistic of 2.45. Among these large cap stocks, the alphas for the CEI strategy relative to the five most common factor models range from 24 to 29 bps/month with t-statistics between 2.70 and 3.30.

5 How does CEI perform relative to the zoo?

Figure 2 puts the performance of CEI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CEI strategy falls in the distribution. The CEI strategy's gross (net) Sharpe ratio of 0.46 (0.37) is greater than 86% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the CEI strategy (red line).[2] Ignoring trading costs, a $1 invested in the CEI strategy would have yielded $4.17 which ranks the CEI strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a $1 invested in the CEI strategy would have yielded $2.71 which ranks the CEI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CEI relative to those. Panel A shows that the CEI strategy gross alphas fall between the 52 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

set for an investor having access to the Fama-French three-factor (six-factor) model. The CEI strategy has a positive net generalized alpha for five out of the five factor models. In these cases CEI ranks between the 71 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CEI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CEI with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CEI or at least to weaken the power CEI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CEI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEI} CEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CEI. Stocks are finally grouped into five CEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CEI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CEI signal in these Fama-MacBeth regressions exceed 1.86, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on CEI is 0.93.

Similarly, Table 5 reports results from spanning tests that regress returns to the CEI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CEI strategy earns alphas that range from 26-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.57, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CEI trading strategy achieves an alpha of 24bps/month with a t-statistic of 3.34.

7 Does CEI add relative to the whole zoo?

Finally, we can ask how much adding CEI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CEI signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CEI grows to \$2730.41.

8 Conclusion

This study provides robust evidence that Capital Expansion Intensity (CEI) serves as an economically meaningful and statistically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that CEI maintains substantial predictive power even after accounting for transaction costs and controlling for established risk factors and closely related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on CEI generates impressive risk-adjusted returns, with an annualized net Sharpe ratio of 0.37 and statistically significant alphas relative to both the Fama-French five-factor model plus momentum (23 bps/month, t-statistic = 3.22) and an extended model including six closely related investment strategies (24 bps/month, t-statistic = 3.34). These results underscore that CEI captures unique information about firm invest-

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CEI is available.

ment behavior not fully explained by existing factors in the literature, including other investment-based measures such as growth in book equity, inventory growth, and changes in net operating assets.

The practical implications of our findings are substantial for both academics and practitioners. For asset managers, CEI presents a viable strategy that survives implementation costs, as evidenced by the modest decline from gross to net Sharpe ratios (0.46 to 0.37). The signal's robustness to multiple factor models suggests it could serve as a valuable addition to multi-factor investment strategies. From a theoretical perspective, the persistence of CEI's predictive power challenges our understanding of market efficiency and highlights the importance of firms' capital allocation decisions in asset pricing.

Several limitations of this study warrant consideration. First, while we follow best practices in accounting for transaction costs, actual implementation costs may vary depending on market conditions and execution capabilities. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of CEI's predictive power to international markets remains an open question. Third, the economic mechanisms driving the CEI effect require further investigation to distinguish between risk-based and behavioral explanations.

Future research should explore several promising avenues. International evidence would help establish whether the CEI effect represents a global phenomenon or is specific to U.S. markets. Time-series variation in the CEI premium could provide insights into its economic drivers and relationship with macroeconomic conditions. Additionally, examining the interaction between CEI and corporate governance characteristics might reveal when capital expansion decisions are more likely to destroy shareholder value. Finally, investigating the real investment outcomes of high versus low CEI firms would enhance our understanding of whether the return predictability stems from systematic mispricing of investment efficiency or compensation for bearing specific risks associated with capital expansion activities.

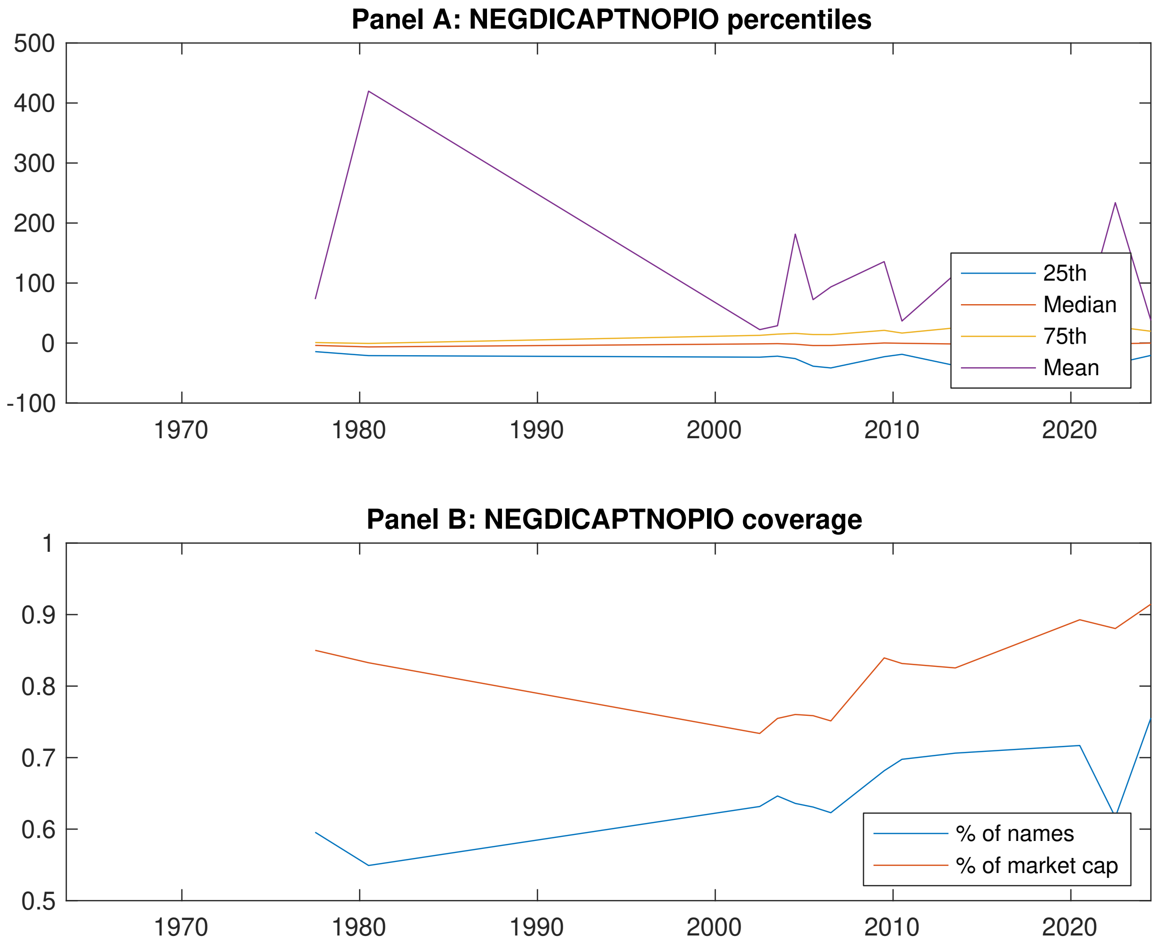


Figure 1: Times series of CEI percentiles and coverage.
This figure plots descriptive statistics for CEI. Panel A shows cross-sectional percentiles of CEI over the sample. Panel B plots the monthly coverage of CEI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CEI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CEI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.56	0.57	0.65	0.73	0.25
	[2.56]	[3.31]	[3.82]	[4.11]	[4.00]	[3.57]
α_{CAPM}	-0.17	-0.02	0.07	0.12	0.11	0.28
	[-3.54]	[-0.44]	[1.40]	[2.26]	[2.01]	[3.90]
α_{FF3}	-0.12	0.02	0.05	0.08	0.15	0.26
	[-2.66]	[0.43]	[1.03]	[1.69]	[2.79]	[3.69]
α_{FF4}	-0.10	0.03	0.04	0.07	0.16	0.26
	[-2.34]	[0.78]	[0.79]	[1.31]	[2.98]	[3.62]
α_{FF5}	-0.10	0.03	-0.03	-0.02	0.19	0.30
	[-2.33]	[0.57]	[-0.66]	[-0.47]	[3.63]	[4.10]
α_{FF6}	-0.10	0.04	-0.03	-0.02	0.20	0.30
	[-2.12]	[0.83]	[-0.69]	[-0.52]	[3.72]	[4.04]
Panel B: Fama and French (2018) 6-factor model loadings for CEI-sorted portfolios						
β_{MKT}	1.05	0.98	0.92	1.00	1.02	-0.04
	[97.59]	[90.59]	[86.05]	[91.31]	[78.63]	[-2.17]
β_{SMB}	0.07	-0.07	-0.11	-0.09	0.06	-0.02
	[4.79]	[-4.25]	[-6.88]	[-5.84]	[3.12]	[-0.65]
β_{HML}	-0.11	-0.04	0.01	-0.07	-0.10	0.00
	[-5.11]	[-2.13]	[0.69]	[-3.36]	[-4.12]	[0.12]
β_{RMW}	0.03	0.05	0.12	0.06	-0.10	-0.14
	[1.66]	[2.18]	[5.63]	[2.84]	[-4.02]	[-3.98]
β_{CMA}	-0.12	-0.11	0.17	0.44	-0.05	0.07
	[-4.02]	[-3.70]	[5.66]	[14.10]	[-1.36]	[1.47]
β_{UMD}	-0.01	-0.02	0.00	0.00	-0.01	0.00
	[-1.05]	[-1.56]	[0.22]	[0.31]	[-0.85]	[0.02]
Panel C: Average number of firms (n) and market capitalization (me)						
n	550	454	441	487	584	
me ($\$10^6$)	1517	1780	2236	1625	1516	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CEI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25	0.28	0.26	0.26	0.30	0.30
			[3.57]	[3.90]	[3.69]	[3.62]	[4.10]	[4.04]
Quintile	NYSE	EW	0.40	0.41	0.37	0.33	0.37	0.34
			[7.64]	[7.66]	[7.50]	[6.67]	[7.34]	[6.72]
Quintile	Name	VW	0.24	0.26	0.26	0.26	0.30	0.30
			[3.19]	[3.51]	[3.45]	[3.37]	[3.94]	[3.83]
Quintile	Cap	VW	0.18	0.23	0.21	0.19	0.24	0.22
			[2.70]	[3.36]	[3.10]	[2.70]	[3.49]	[3.15]
Decile	NYSE	VW	0.24	0.27	0.28	0.26	0.31	0.29
			[2.66]	[2.96]	[3.04]	[2.76]	[3.26]	[3.02]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.21	0.22	0.21	0.21	0.23	0.23
			[2.92]	[3.10]	[2.88]	[2.89]	[3.22]	[3.22]
Quintile	NYSE	EW	0.19	0.19	0.15	0.14	0.12	0.11
			[3.14]	[3.07]	[2.60]	[2.33]	[2.13]	[1.95]
Quintile	Name	VW	0.19	0.21	0.20	0.20	0.23	0.23
			[2.56]	[2.72]	[2.64]	[2.63]	[3.01]	[2.99]
Quintile	Cap	VW	0.14	0.18	0.16	0.15	0.18	0.17
			[2.11]	[2.66]	[2.36]	[2.15]	[2.73]	[2.56]
Decile	NYSE	VW	0.19	0.20	0.21	0.20	0.23	0.22
			[2.08]	[2.22]	[2.26]	[2.13]	[2.42]	[2.31]

Table 3: Conditional sort on size and CEI

This table presents results for conditional double sorts on size and CEI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CEI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CEI and short stocks with low CEI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

	CEI Quintiles					CEI Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.45	0.86	0.92	0.97	0.78	0.33	0.31	0.30	0.23	0.30	0.24
	[1.81]	[3.53]	[3.86]	[3.76]	[3.00]	[4.10]	[3.88]	[3.70]	[2.82]	[3.57]	[2.85]
(2)	0.69	0.70	0.90	0.88	0.91	0.22	0.23	0.21	0.25	0.24	0.28
	[2.87]	[3.11]	[4.01]	[3.96]	[3.84]	[2.49]	[2.63]	[2.40]	[2.79]	[2.64]	[2.97]
(3)	0.63	0.74	0.76	0.84	0.89	0.27	0.27	0.24	0.21	0.20	0.18
	[2.90]	[3.55]	[3.83]	[4.26]	[4.11]	[3.31]	[3.33]	[2.92]	[2.57]	[2.43]	[2.22]
(4)	0.63	0.65	0.70	0.71	0.85	0.22	0.22	0.17	0.15	0.15	0.14
	[3.11]	[3.50]	[3.80]	[3.87]	[4.15]	[3.02]	[2.96]	[2.37]	[2.04]	[2.05]	[1.83]
(5)	0.42	0.50	0.56	0.66	0.63	0.22	0.26	0.25	0.24	0.29	0.29
	[2.25]	[2.91]	[3.75]	[4.26]	[3.63]	[2.45]	[3.00]	[2.85]	[2.70]	[3.30]	[3.15]

Panel B: Portfolio average number of firms and market capitalization

	CEI Quintiles: Average n					CEI Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	258	259	260	256	254	24	24	22	21	22
(2)	79	79	79	79	78	42	42	43	42	43
(3)	61	61	61	60	60	78	78	77	78	78
(4)	55	55	55	55	55	182	177	181	178	177
(5)	52	52	52	52	51	1116	1324	1860	1507	1258

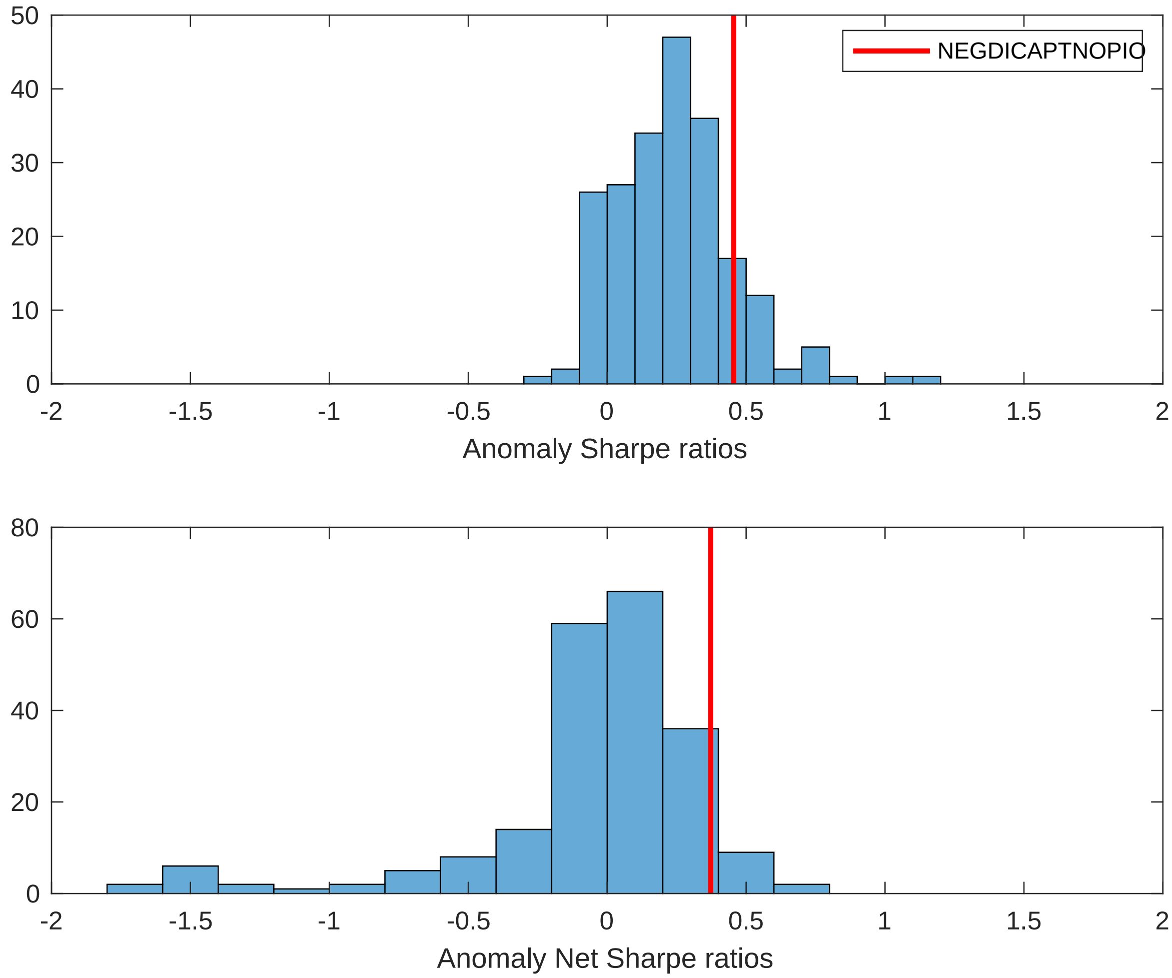


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CEI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

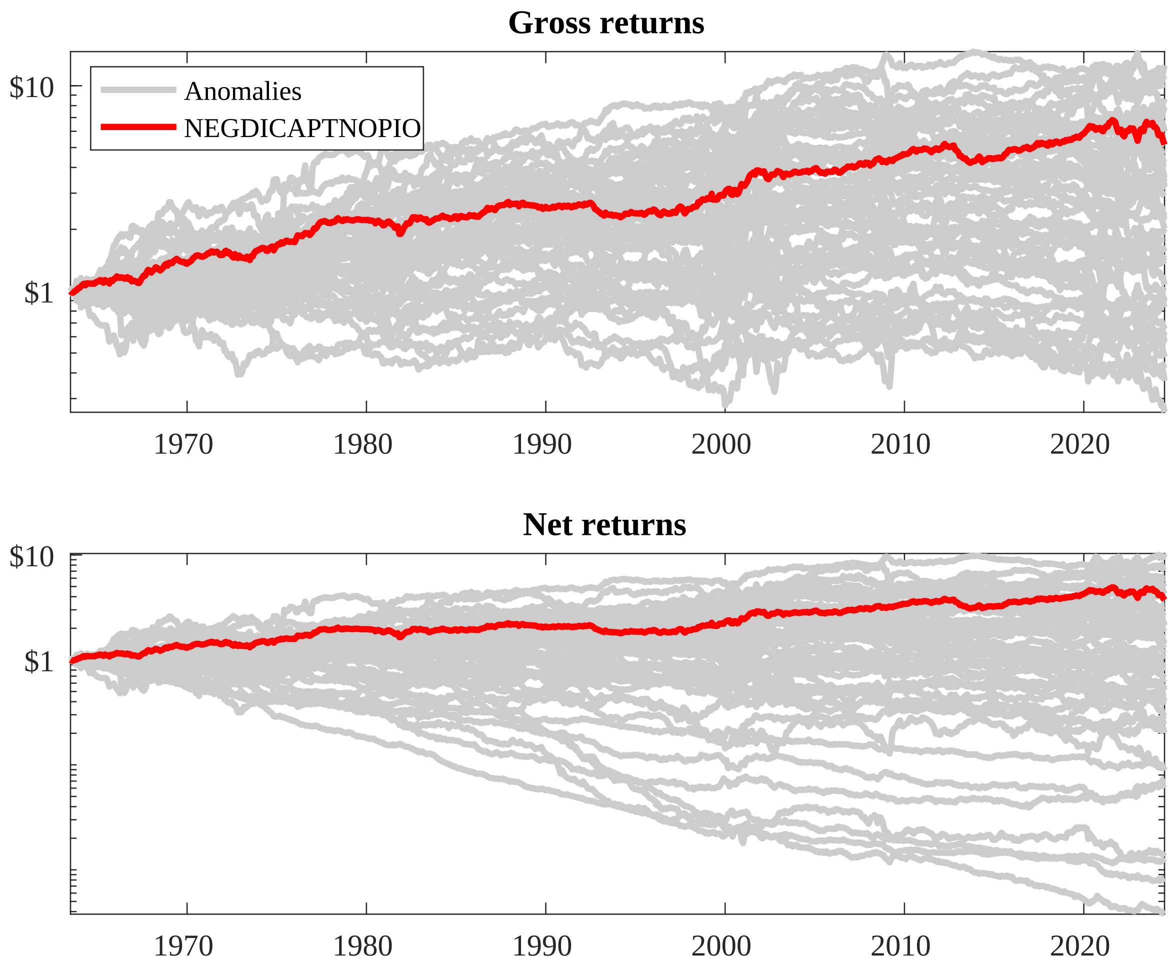
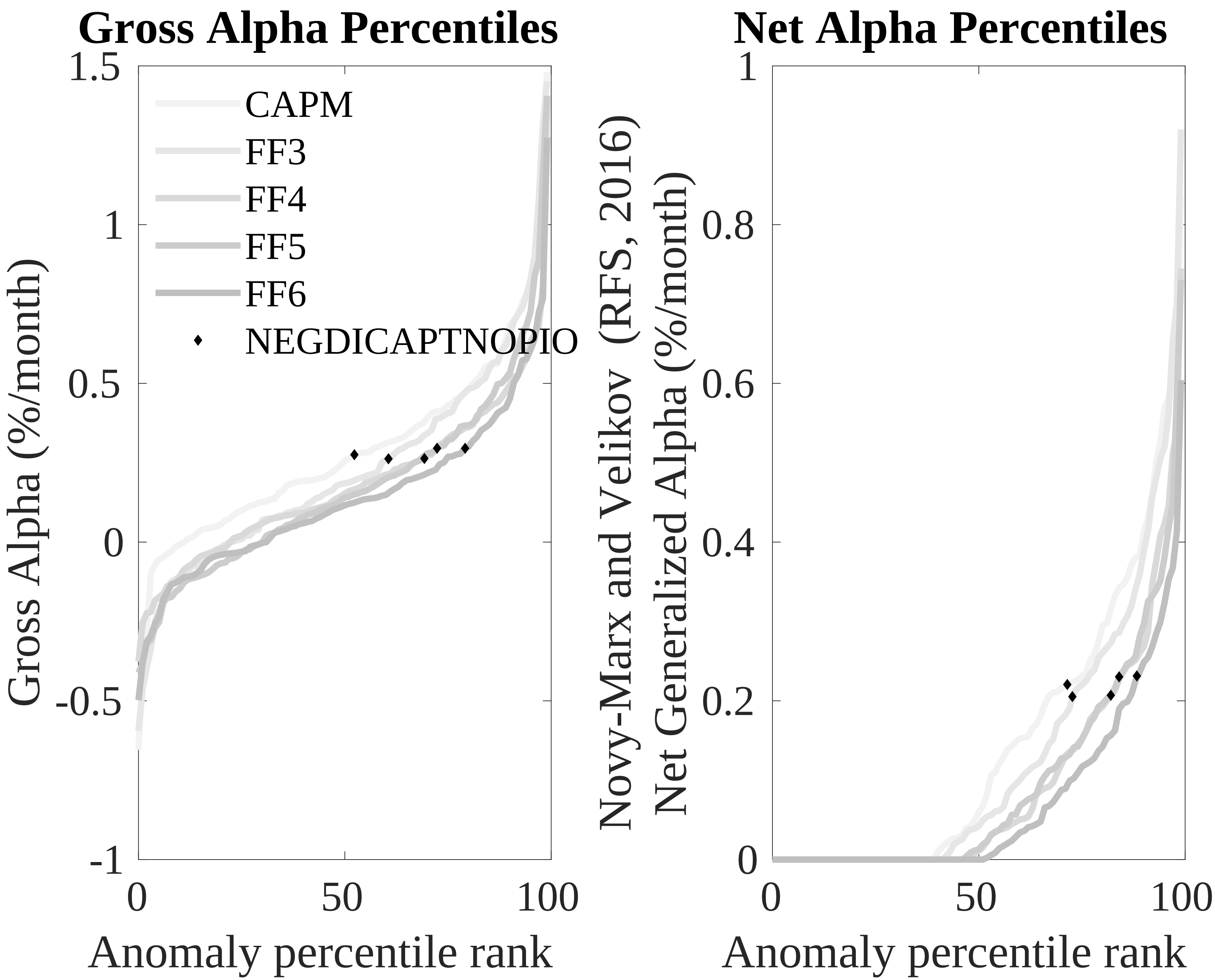


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CEI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDICAPTNOPIO
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
1
0.8
0.6
0.4
0.2
0
0
50
100
0
50
100

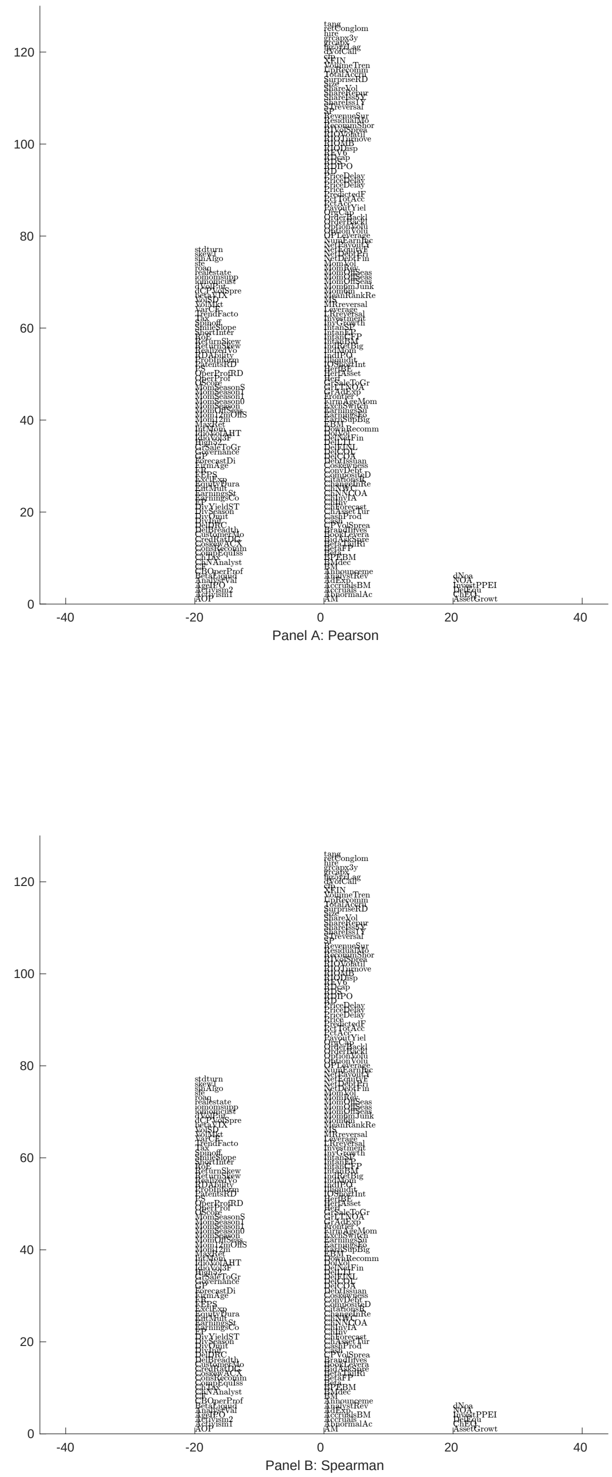


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with CEI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

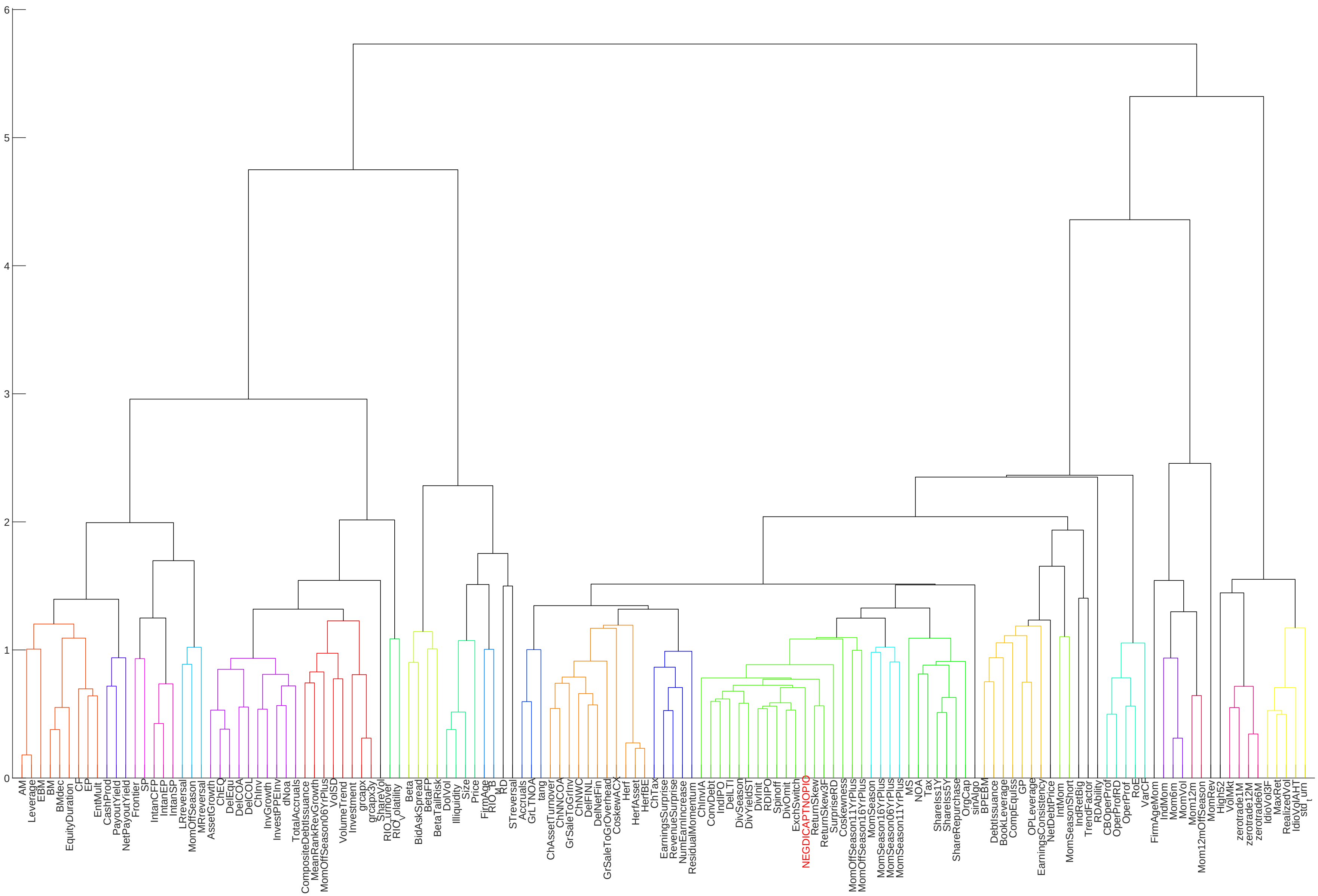


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

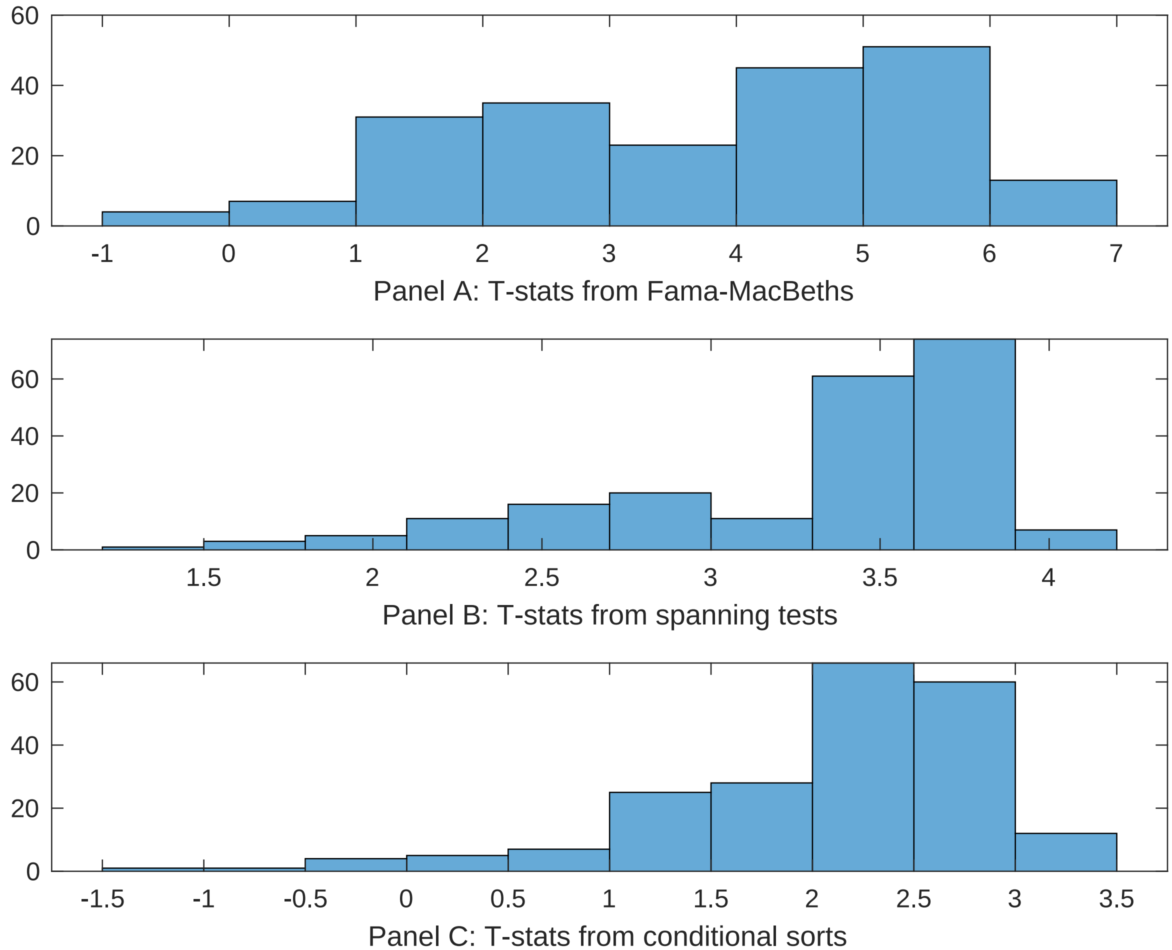


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of CEI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CEI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CEI} CEI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CEI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CEI. Stocks are finally grouped into five CEI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CEI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CEI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CEI} CEI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17	0.13	0.12	0.13	0.13	0.17	0.15
	[6.82]	[6.03]	[5.58]	[5.67]	[5.82]	[7.02]	[5.03]
CEI	0.37	0.24	0.41	0.34	0.16	0.29	0.95
	[4.24]	[2.89]	[4.56]	[3.22]	[1.86]	[3.46]	[0.93]
Anomaly 1	0.41						0.81
	[3.57]						[0.72]
Anomaly 2		0.16					0.43
		[7.83]					[1.43]
Anomaly 3			0.29				0.28
			[5.58]				[0.41]
Anomaly 4				0.49			0.94
				[8.13]			[1.30]
Anomaly 5					0.13		0.61
					[9.13]		[2.43]
Anomaly 6						0.83	0.19
						[6.85]	[1.08]
# months	720	732	732	732	720	732	720
$\bar{R}^2$(%)	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CEI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CEI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Growth in book equity, change in ppe and inv/assets, Inventory Growth, Inventory Growth, change in net operating assets, Net Operating Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.28	0.29	0.26	0.29	0.28	0.26	0.24
	[3.84]	[3.93]	[3.59]	[3.91]	[3.80]	[3.57]	[3.34]
Anomaly 1	12.12						10.23
	[3.06]						[2.53]
Anomaly 2		7.91					2.73
		[2.31]					[0.70]
Anomaly 3			12.85				12.55
			[3.57]				[2.76]
Anomaly 4				6.09			-2.53
				[2.06]			[-0.67]
Anomaly 5					3.80		-5.74
					[0.92]		[-1.26]
Anomaly 6						9.68	8.90
						[3.14]	[2.70]
mkt	-3.43	-3.93	-3.42	-3.91	-3.81	-4.18	-3.45
	[-1.96]	[-2.25]	[-1.96]	[-2.24]	[-2.17]	[-2.39]	[-1.98]
smb	-1.15	-1.06	0.81	-0.34	-0.70	0.80	1.51
	[-0.45]	[-0.42]	[0.32]	[-0.13]	[-0.28]	[0.31]	[0.58]
hml	-0.20	0.43	0.44	1.00	1.00	-1.32	-2.94
	[-0.06]	[0.13]	[0.13]	[0.29]	[0.30]	[-0.38]	[-0.85]
rmw	-12.84	-13.46	-11.22	-12.70	-13.39	-14.43	-12.08
	[-3.77]	[-3.95]	[-3.26]	[-3.70]	[-3.91]	[-4.23]	[-3.50]
cma	-5.11	0.56	-1.78	1.64	3.80	6.48	-7.38
	[-0.82]	[0.10]	[-0.32]	[0.30]	[0.65]	[1.31]	[-1.06]
umd	-0.19	0.01	-0.75	-0.63	-0.05	-0.32	-0.86
	[-0.11]	[0.01]	[-0.43]	[-0.35]	[-0.03]	[-0.19]	[-0.49]
# months	732	732	732	732	732	732	732
$\bar{R}^2$(%)	4	4	5	4	3	4	6

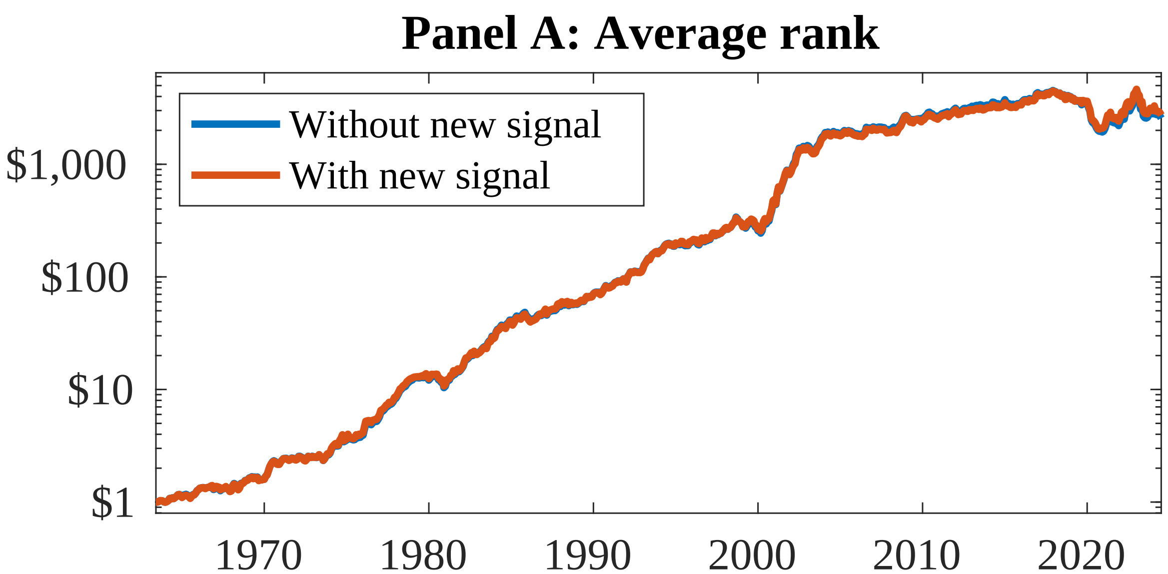


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CEI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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